nature catalysis

SUPPLEMENTARY INFORMATION

https://doi.org/10.1038/s41929-020-0459-4

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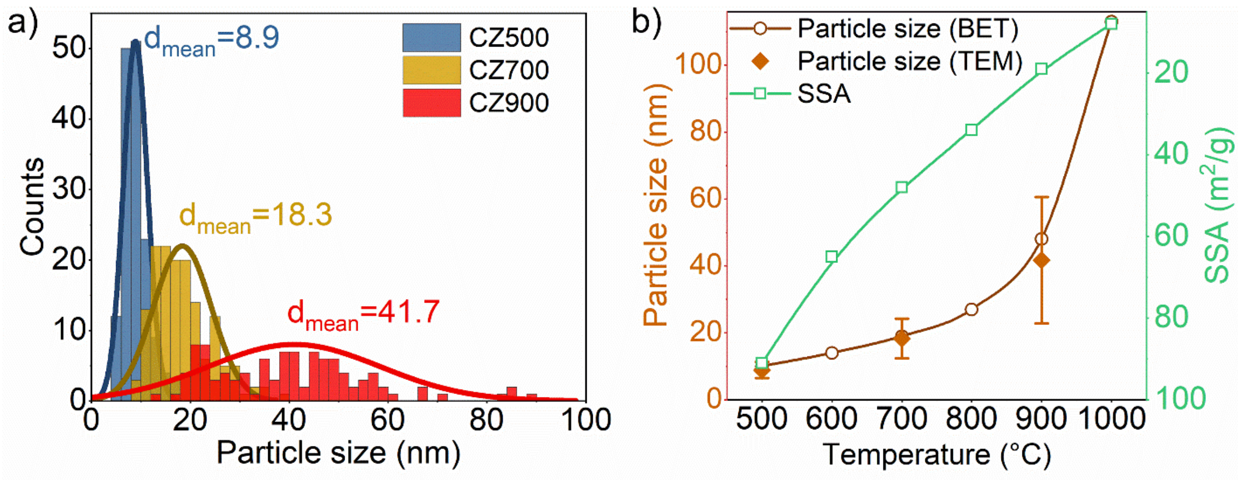
Supplementary information

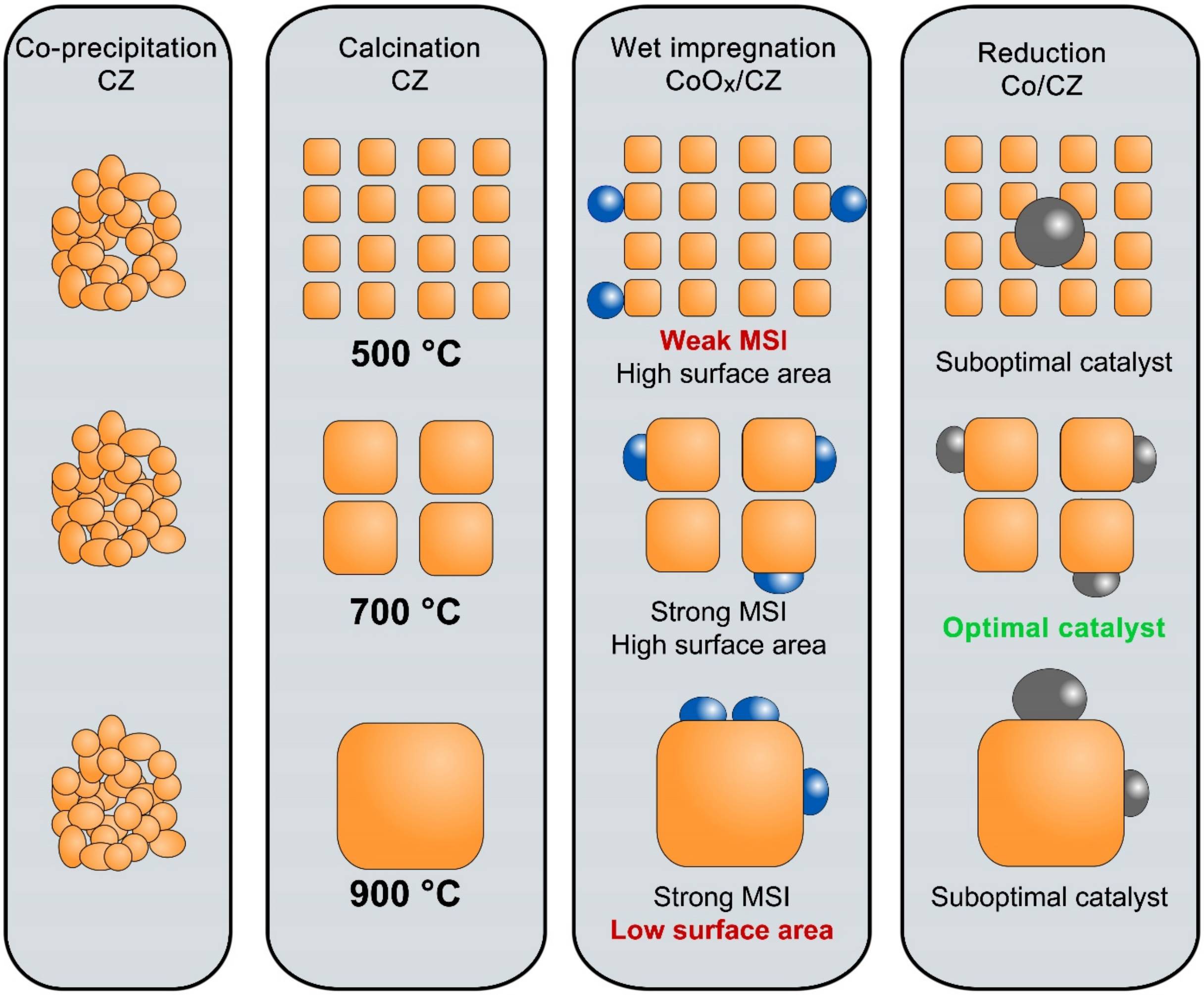


Supplementary Figure 1. a) Particle size distribution and mean particle size obtained from TEM images for CZ500, CZ700 and CZ900; b) Comparison of support particle size obtained by BET and TEM. Error bars represent the standard deviation in support mean particle sizes determined by TEM.

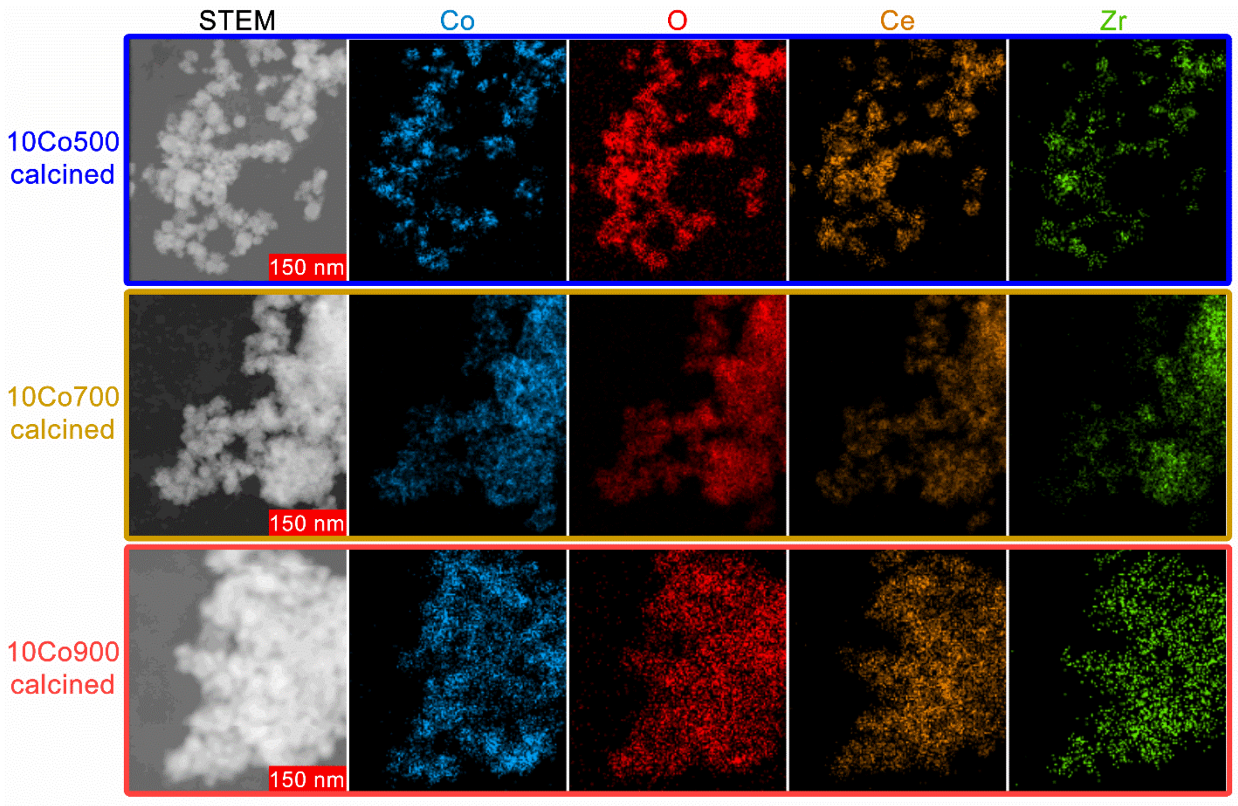
Supplementary Table 1. BET surface area and particle size of the supports.

Support	BET, m^2/g	Particle size[a], nm
CZ500	91	10
CZ600	65	14
CZ700	48	19
CZ800	34	27
CZ900	19	48
CZ1000	8	113
CZ SA	54	17
TiO_2 P25	58	25

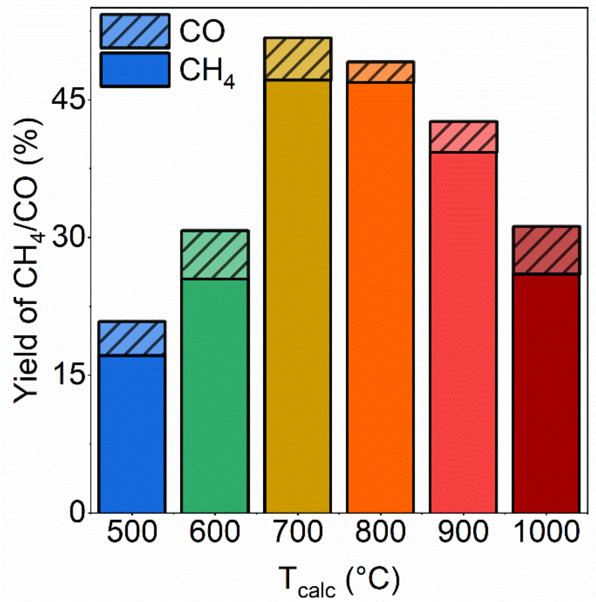
[a] estimated from BET surface area with assumption that particles are non-porous and spherical. Density of ceria-zirconia – 6.61 g/cm^3. Density of TiO_2 – 4.23 g/cm^3.



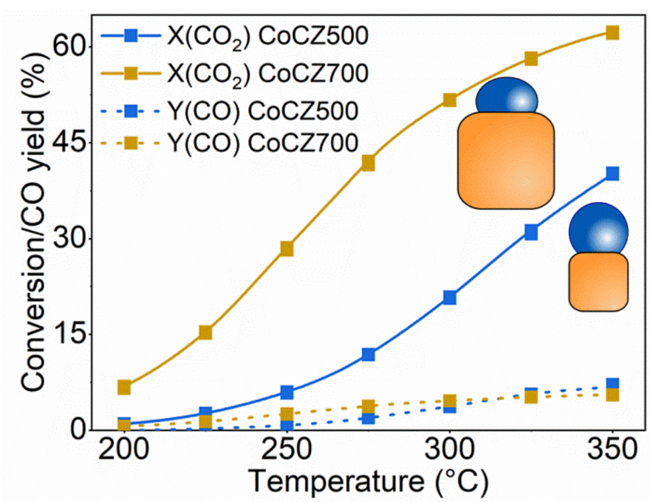
Supplementary Figure 2. Catalysts preparation procedure involving following steps: co-precipitation of ceria-zirconia, calcination of obtained material at different temperatures, impregnation with cobalt precursor and calcination at 350°C, reduction at 500°C.



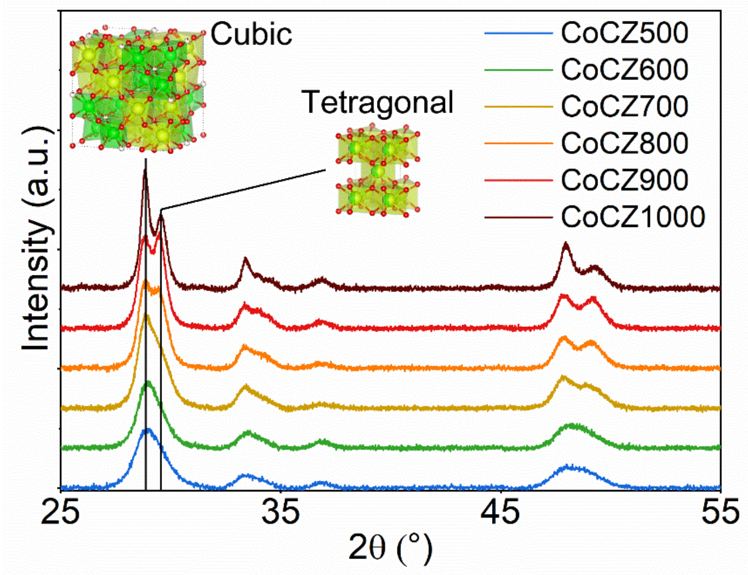
Supplementary Figure 3. STEM images and EDX mapping (Co, O, Ce, Zr) of calcined cobalt catalyst: CoCZ500, CoCZ700 and CoCZ900.



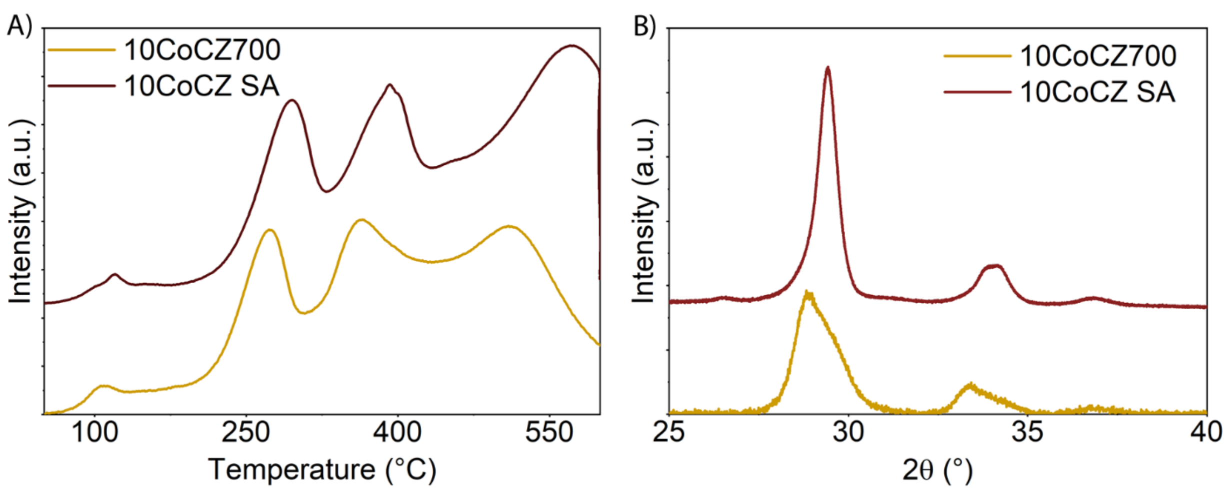
Supplementary Figure 4. Catalytic performance at 300 °C of cobalt based catalysts supported on CZ as a function of CZ calcination temperature.



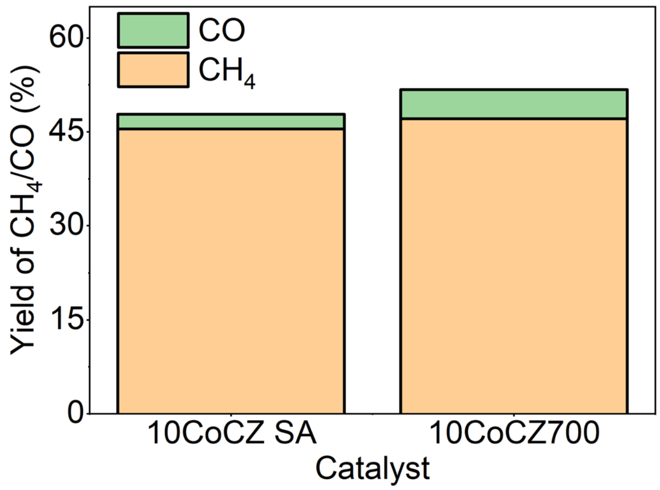
Supplementary Figure 5. CO_2 conversion and CO yield as a function of reaction temperature for CoCZ500 and CoCZ700.



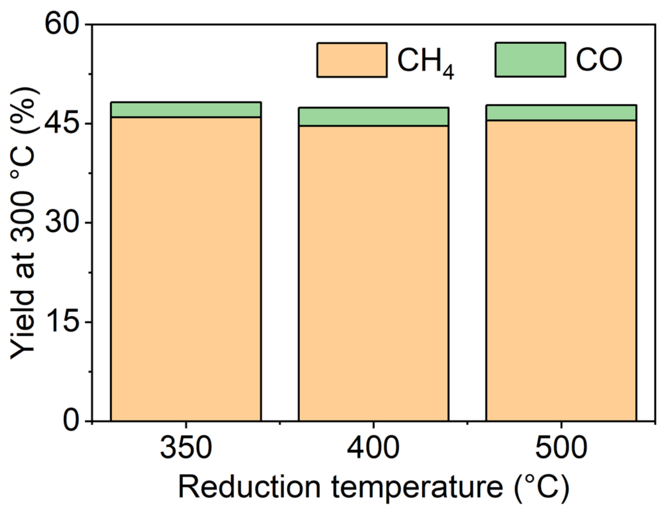
Supplementary Figure 6. XRD patterns of calcined CZ samples.



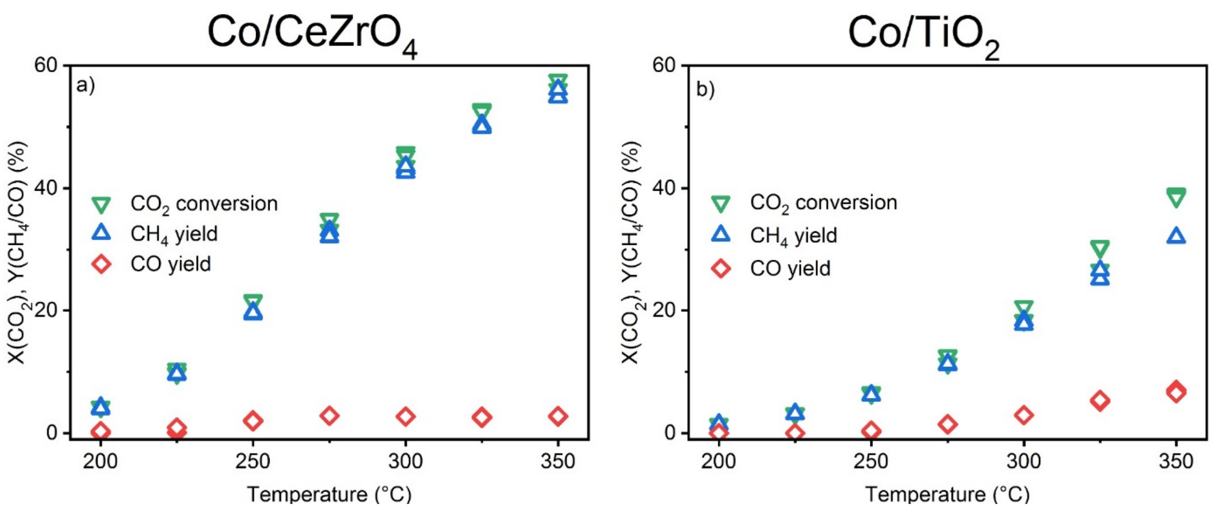
Supplementary Figure 7. Comparison of CoCZ700 and CoCZ SA: A) TPR profiles; B) XRD patterns.



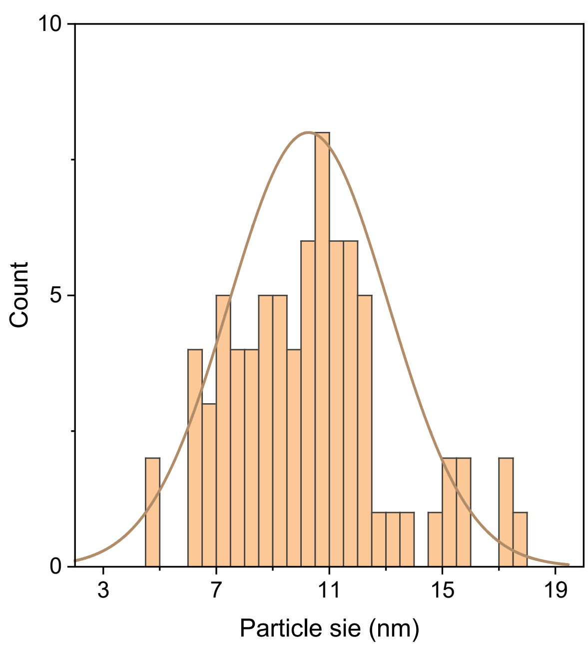
Supplementary Figure 8. Influence of crystal phase on catalytic activity: catalytic activity at 300 °C of CoCZ700 and CoCZ SA.



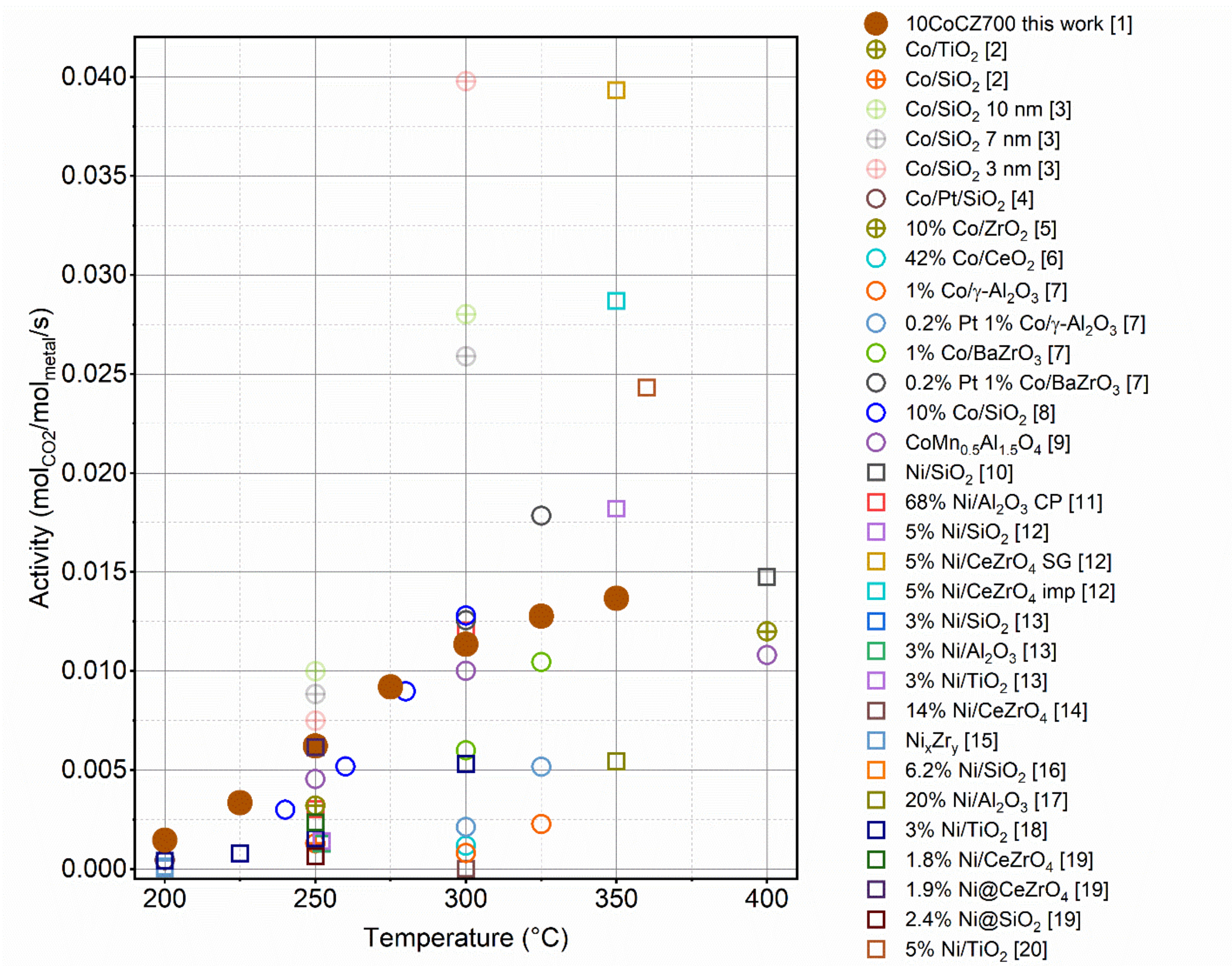
Supplementary Figure 9. Yield of CO and CH_4 in CO_2 methanation over 10CoCZ-SA catalysts reduced at different temperature. Conditions: 1 bar, 300 °C, 50 mg of catalyst, 5 % CO_2, 20 % H_2, He balance, total flow of 50 mL/min.



Supplementary Figure 10. Catalytic performance of 10Co/CZ (commercial) and 8.25Co/TiO_2 (P25).



Supplementary Figure 11. Particle size distribution of cobalt for calcined 8.25Co/TiO_2.



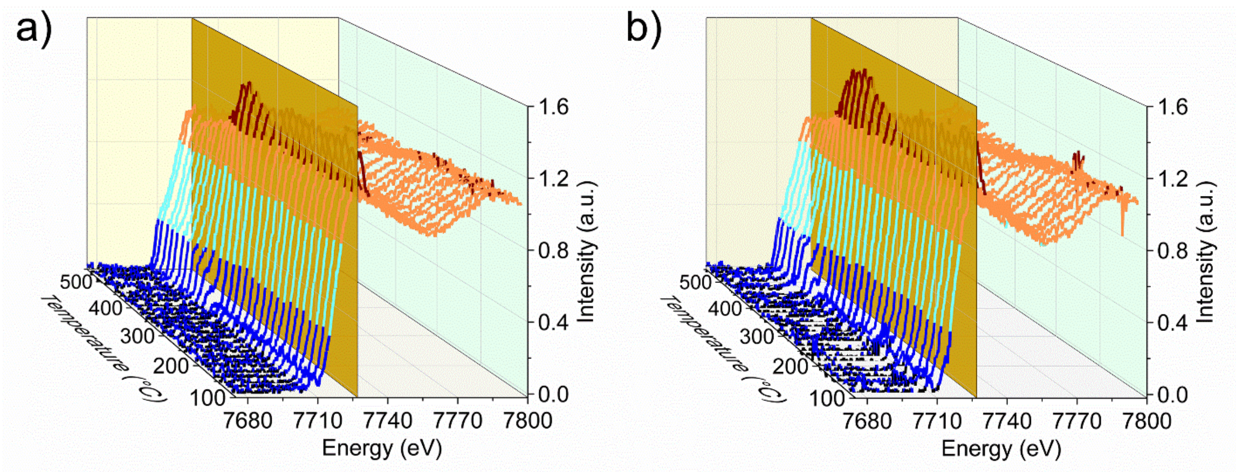
Supplementary Figure 12. Activity of catalysts in CO_2 methanation normalized to the total amount of metal at different temperatures. ○ – cobalt-based catalysts; ⊕ – cobalt-based catalysts at elevated pressure; □ – nickel-based catalysts. Numbers in the legend correspond to the position of catalysts in the table below.

Supplementary Table 2. Activity of nickel and cobalt-based catalysts in CO_2 methanation.

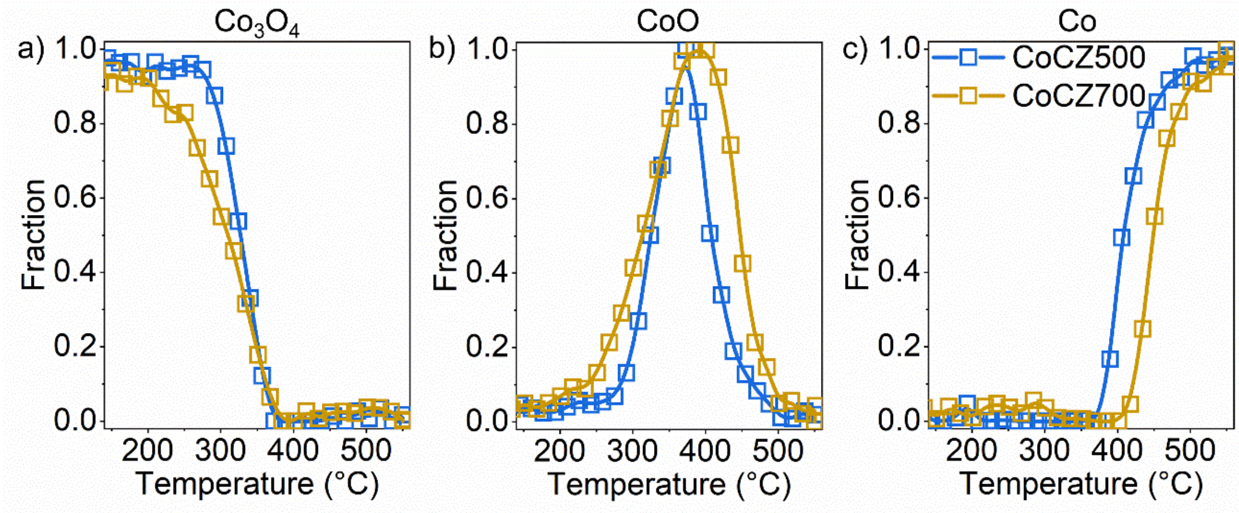
	Catalyst	CO_2 conc., %	Conditions Remarks	T_{reac}, °C	Specific activity, $mol_{CO2}/mol_{metal}/s$	Ref.
1	10CoCZ700	5	1 bar	200 225 250 275 300 325	1.46E-03 3.35E-03 6.22E-03 9.19E-03 1.13E-02 1.28E-02	This work

				350	1.37E-02	
2	Co/TiO_2	25	Oxidized, 5 bar	250	3.20E-03	1
	Co/SiO_2		Reduced, 5 bar	250	1.30E-03	
3	Co/SiO_2 10nm	22.2	6 bar	250 300	1.00E-01 2.80E-01	2
	Co/SiO_2 7 nm			250 300	6.20E-02 1.81E-01	
	Co/SiO_2 3nm			250 300	2.25E-02 1.19E-01	
4	$Co/Pt/SiO_2$	22.2	1 bar	200	4.99E-04	3
5	10% Co/ZrO_2	20	30 bar	400	1.20E-02	4
6	42% Co/CeO_2	10	1 bar	300	1.18E-03	5
7	1%Co/γ-Al_2O_3	3	1 bar	300 325	8.13E-04 2.27E-03	6
	0.2%Pt1%Co/γ-Al_2O_3			300 325	2.12E-03 5.17E-03	
	1%Co/$BaZrO_3$			300 325	5.99E-03 1.05E-02	
	0.2%Pt 1%Co/$BaZrO_3$			300 325	1.26E-02 1.78E-02	
8	10%Co/SiO_2	10	1 bar	240 260 280 300	3.00E-03 5.18E-03 8.98E-03 1.28E-02	7
9	$CoMn_{0.5}Al_{1.5}O_4$	20	1 bar	250 300 400	4.55E-03 1.00E-02 1.08E-02	8
10	Ni/SiO_2	10	1 bar	400	1.48E-02	9
11	68%Ni/Al_2O_3 CP	19	2 bar	250 300	3.01E-03 1.21E-02	10
12	5%Ni/SiO_2 5%Ni/$CeZrO_4$ SG 5%Ni/$CeZrO_4$ imp	16.4	1 bar	350 350 350	1.82E-02 3.93E-02 2.87E-02	11
13	3%Ni/SiO_2 3%Ni/Al_2O_3 3%Ni/TiO_2	1	1 bar	252 252 252	1.40E-03 1.29E-03 1.43E-03	12
14	14%Ni/$CeZrO_4$	20	1 bar	300	1.48E-05	13
15	$Ni_{70}Zr_{30}$ $Ni_{60}Zr_{40}$ $Ni_{50}Zr_{50}$ $Ni_{40}Zr_{60}$ $Ni_{30}Zr_{70}$ 4%Ni/ZrO_2	20	1 bar	200 200 200 200 200 200	3.46E-05 7.19E-05 1.02E-04 7.06E-05 4.50E-05 1.24E-04	14
16	6.2%Ni/SiO_2	20	1 bar	450	5.32E-03	15

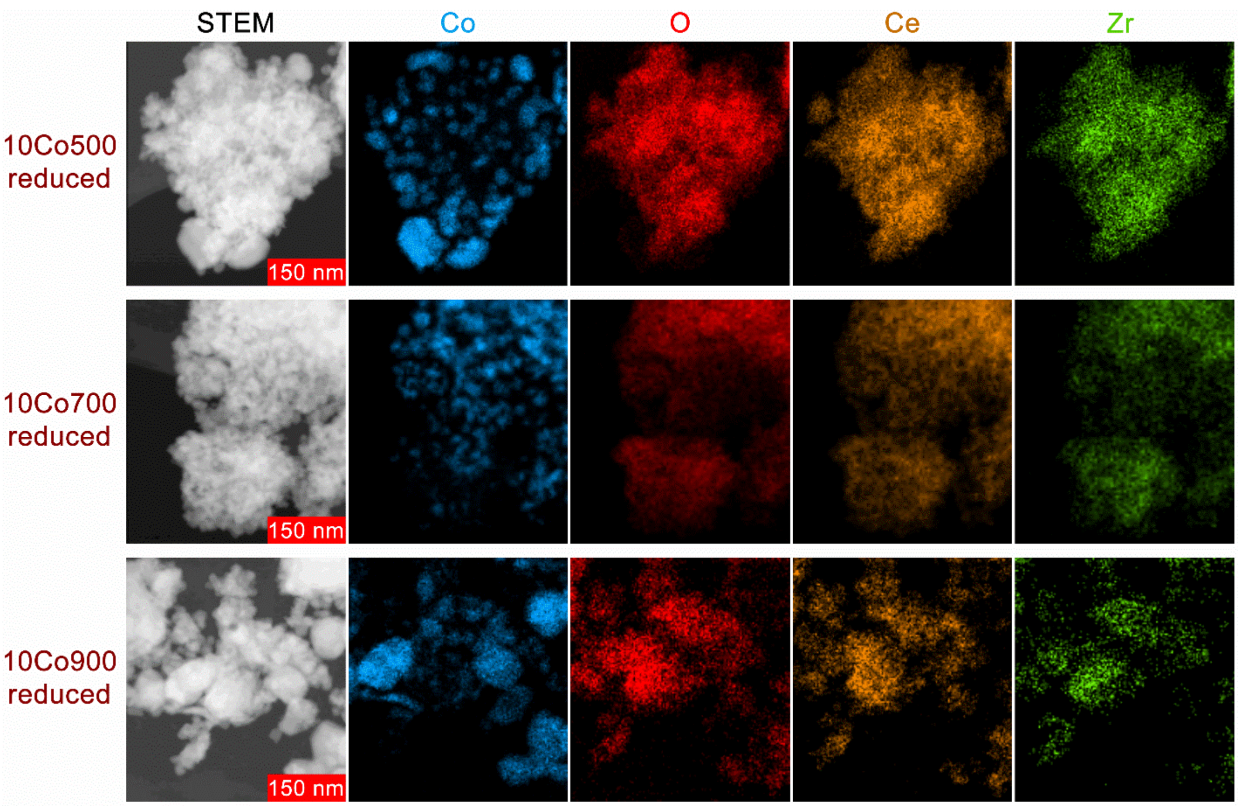
17	20%Ni/Al_2O_3	20	1 bar	350	5.43E-03	[16]
18	3 % Ni/TiO_2	5	1 bar	200 225 250 300	4.10E-04 8.00E-04 1.56E-03 5.31E-03	[17]
19	1.8 % Ni/$CeZrO_4$ 1.9 % Ni@$CeZrO_4$ 2.4 % Ni@SiO_2	5	1 bar	250	2.35E-03 6.15E-03 0.64E-03	[18]
20	5 % Ni/TiO_2	19	1 bar	360	2.43E-02	[19]



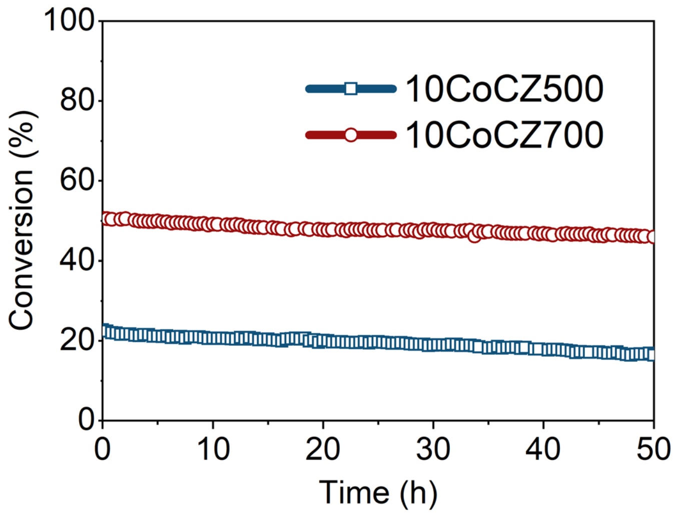
Supplementary Figure 13. XANES spectra obtained during reduction of CoCZ500 (a) and CoCZ700 (b).



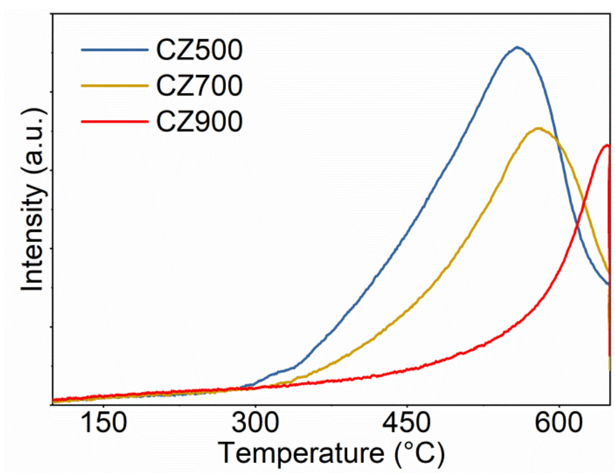
Supplementary Figure 14. Results of linear combination fitting of XANES spectra performed for CoCZ500 and CoCZ700. a) Co_3O_4; b) CoO; c) Co^0.



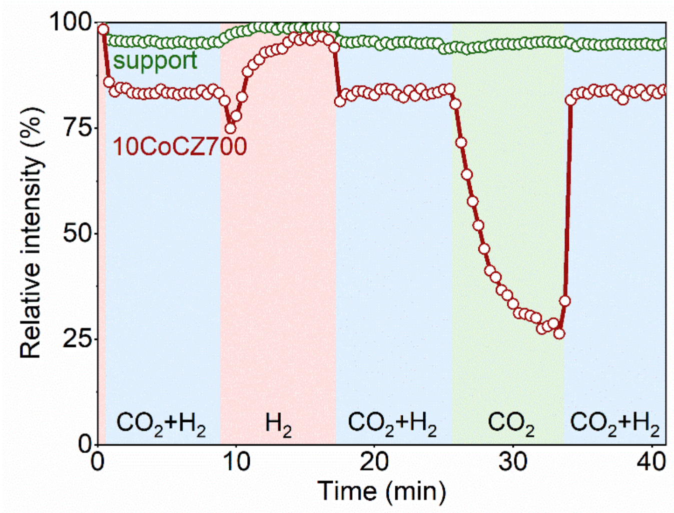
Supplementary Figure 15. STEM images and EDX mapping (Co, O, Ce, Zr) of reduced catalysts: CoCZ500, CoCZ700 and CoCZ900.



Supplementary Figure 16. Stability of CoCZ catalysts in CO_2 hydrogenation. Conditions: 1 bar, 300 °C, 50 mg of catalyst, 5 % CO_2, 20 % H_2, He balance, total flow of 50 mL/min.

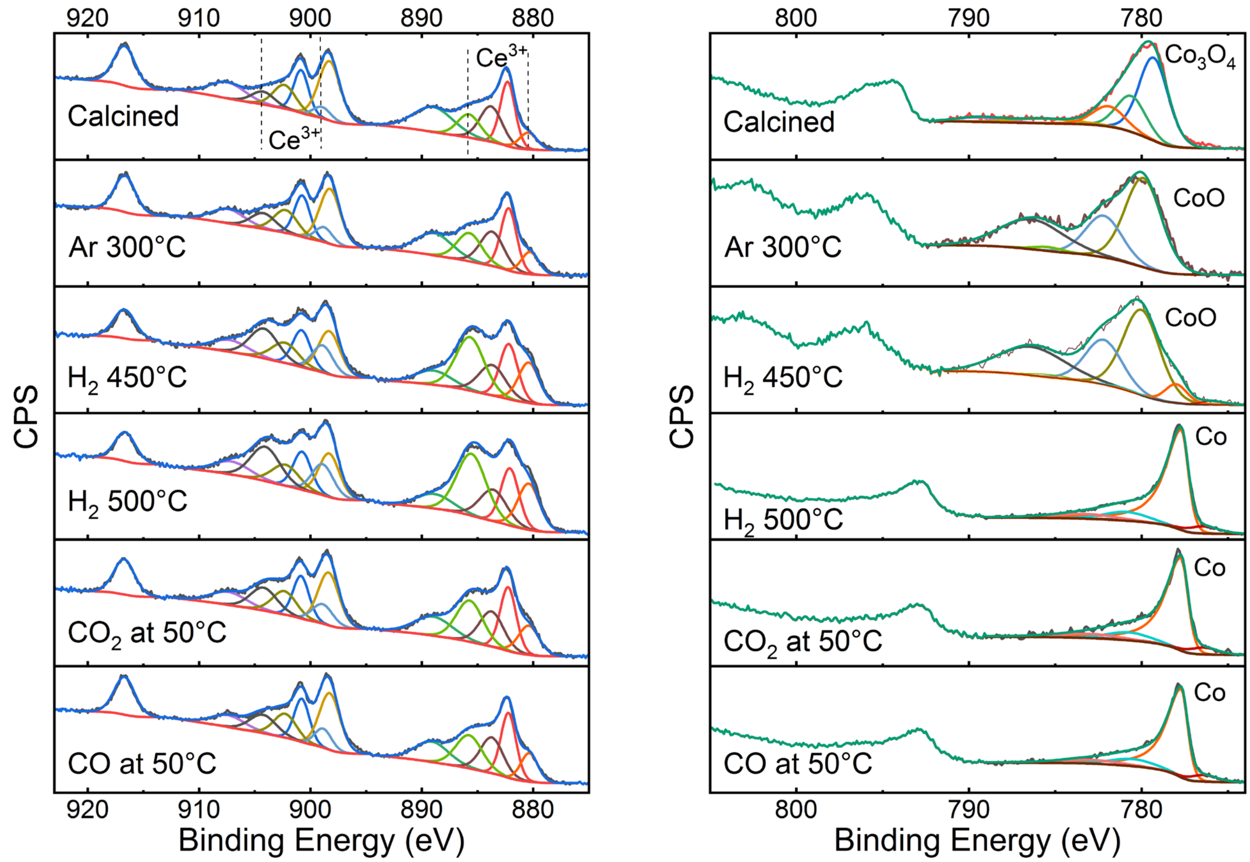


Supplementary Figure 17. TPR profiles of bare ceria-zirconia calcined at 500, 700 and 900 °C.

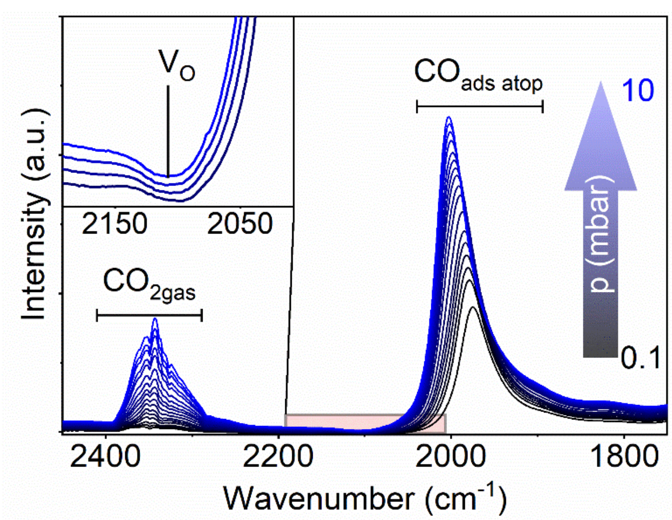


Supplementary Figure 18. Quantification of Ce^{3+} electronic transition band followed by FTIR during transient experiments at 300 °C. Conditions: reduction time 4 h, 1 bar, 300 °C, 12 mg of catalyst (pellet), 2.5 % CO_2, 10 % H_2, He balance, total flow of 200 mL/min.

In order to compare the formation and evolution of oxygen vacancies for bare CZ700 support and CoCZ700 catalyst under reaction conditions (300 °C) we performed transient in situ IR experiments. In these tests the feed composition in the in situ cell was swiftly switched from H_2 to CO_2 + H_2, then to H_2, then to CO_2 + H_2, then to CO_2 and then finally to CO_2 + H_2 again (see Supplementary Figure 18). During all these switches we followed the extent of ceria reduction by quantifying the area of the band at 2115 cm^{-1}. Bare support, reduced at 500 °C, demonstrates negligible changes in Ce^{3+} concentration. Re-oxidation of surface Ce^{3+} species after switching to both the reaction CO_2 + H_2 mixture and pure CO_2 amounts to ca. 5 %. In turn, CoCZ700 catalyst with enhanced MSI demonstrated fast dynamic changes during the transient experiments and significant re-oxidation of the ceria support in CO_2 atmosphere and reduction in H_2 atmosphere.

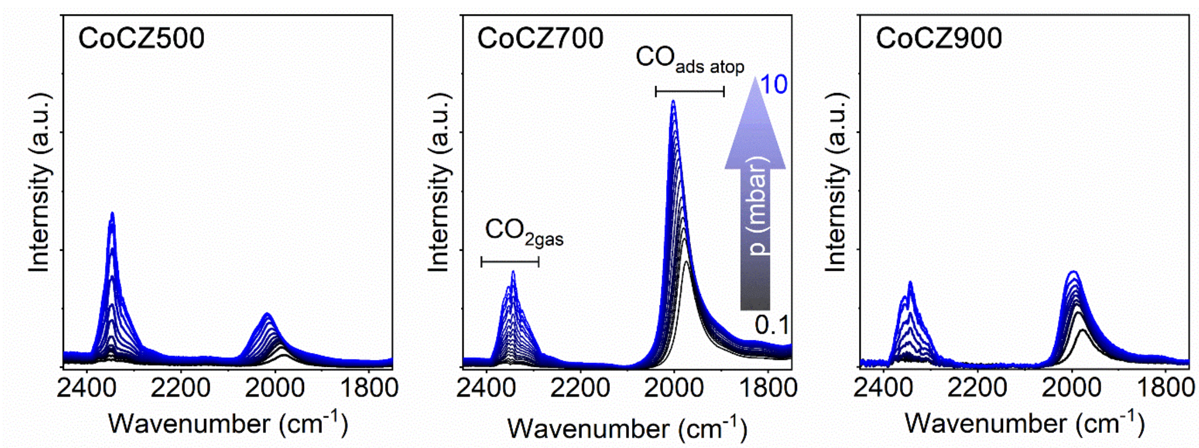


Supplementary Figure 19. NAP-XPS spectra of CoCZ700.



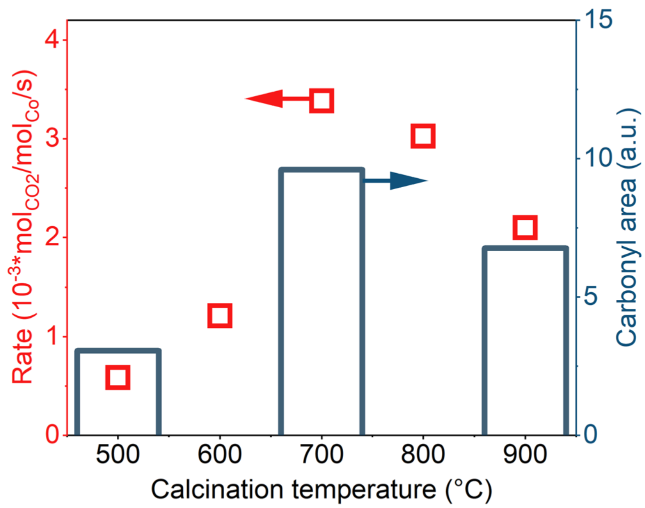
Supplementary Figure 20. Difference IR spectra obtained by exposing the 10Co700 to increasing partial pressure of CO_2 at 50 °C.

Partial re-oxidation of the surface Ce^{3+} is mainly caused by the oxygen spillover from metallic cobalt/cobalt-ceria interface to the ceria surface. In order to obtain a direct evidence of CO_2 dissociation on metallic cobalt sites/cobalt-ceria interface, we exposed CoCZ700 to a different partial pressure of CO_2 (0.1 – 10 mbar) at 50 °C. The presence of intense band of carbonyl species on metallic cobalt surface confirms that CO_2 dissociation takes place and explains the partial oxidation of surface Ce^{3+} observed by NAP-XPS analysis. Moreover, the inset on the Supplementary Figure 20 demonstrates a negative band at 2114 cm^{-1}, pointing at partial re-oxidation of oxygen vacancies upon CO_2 exposure.



Supplementary Figure 21. CO_2 adsorption at 50 °C on reduced catalysts samples followed by IR spectroscopy. Conditions: 50 °C, 12 mg of catalyst, 0.1 – 10 mbar partial pressure of CO_2.

For these experiments we reduced the CoCZ500, CoCZ700 and CoCZ900 catalysts in an *in situ* IR cell (transmission mode) by hydrogen at 500 °C. Then we evacuated the cell and exposed the catalysts to a CO_2 atmosphere of increasing pressure at 50 °C. The results (Supplementary Figure 21) show two groups of vibrational bands appearing after CO_2 exposure. Bands centred around 2350 cm^{-1} correspond to physisorbed (sharp contribution more pronounced for CoCZ500 with higher surface area)[20] and gas phase CO_2. Bands related to carbonyl groups on Co appear around 2000 cm^{-1}. The formation of carbonyls is a result of vacancy-assisted dissociation of CO_2 or dissociation on the metallic Co surface.[21] In this process CO_2 dissociates, yielding a CO molecule adsorbed on Co, while the O atom fills an oxygen vacancy in the ceria support, resulting in the re-oxidation of Ce^{3+} to Ce^{4+} (also observed by NAP-XPS Figs. 4g and 4h in the main text).



Supplementary Figure 22. Correlation between the conversion and ability to dissociate CO_2 for CoCZ catalysts. Values of carbonyl areas are obtained by integration of the corresponding band observed during the exposure of CoCZ catalysts to 10 mbar of CO_2 at 50 °C.

Supplementary References

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